

MEYSAM ZAREI

M.Sc. Medical Biotechnology (Bioinformatics) | Computational Genomics | Machine Learning in Biology

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PROFESSIONAL SUMMARY

Bioinformatics graduate with hands-on experience designing and executing reproducible computational pipelines for bulk RNA-seq, single-cell transcriptomics, and NGS data analysis, with a focus on cancer genomics. Skilled in applying statistical modelling and supervised machine learning to high-dimensional biological datasets, and in translating complex computational outputs into biologically meaningful insights. Proficient in R and Python; experienced with Bioconductor, DESeq2, edgeR, Scanpy, and Squidpy. Committed to rigorous, reproducible research and effective collaboration between computational and experimental teams.

EDUCATION

M.Sc. Medical Biotechnology — Specialization in Bioinformatics

Oct 2022 – Mar 2025

University of Piemonte Orientale (UPO), Novara, Italy

Thesis: "Functional Annotation Analysis in Lung Cancer via Integration of Transcriptomic Data"

Analysed 76 NSCLC patient samples (20,050 genes); performed differential expression (edgeR), GSEA, and molecular clustering to characterise tumour heterogeneity.

B.Sc. Biology

Oct 2015 – Oct 2019

Payam Noor University, Isfahan, Iran

RESEARCH EXPERIENCE

Bioinformatics Intern | Molecular Biology and Bioinformatics Lab, UPO (CAAD) | Italy Oct 2023 – Mar 2025

- Analysed bulk RNA-seq data from 76 NSCLC patient samples (20,050 genes), applying dual-layer classification (histological vs. molecular clustering) to identify transcriptional signatures driving tumour heterogeneity.
- Identified 382 differentially expressed genes (DEGs) using edgeR ($|\log_2FC| > 1$, $FDR < 0.05$) and performed GSEA against MSigDB gene sets ($|NES| > 2$, $FDR < 0.01$), revealing MYC targets, E2F targets, and cell cycle pathways as key adenocarcinoma drivers.
- Engineered a custom R script to resolve a cross-dataset integration bottleneck, improving pipeline consistency across all 76 samples; validated findings with Metascape and ToppGene on 122 DEGs.
- Collaborated with experimental biologists to interpret computational outputs in the context of tumour biology, producing analysis reports that bridged computational and wet-lab teams.

Laboratory Assistant | Payam Noor University, Department of Biology | Isfahan, Iran

Fall 2018

- Supported molecular biology laboratory operations (~25 hours/week), preparing reagents, calibrating equipment, and assisting students with DNA extraction, agarose gel electrophoresis, and PCR protocols.

RESEARCH PROJECTS

NGS Genomic Variant Analysis Pipeline

May – Jun 2025

- Designed and executed a complete 5-stage NGS pipeline (QC, adapter trimming, alignment, SNP calling, de novo assembly) on a 3.5M-read viral genomic dataset; retained 97.08% of reads post-filtering.
- Resolved a low initial mapping rate (26.98%) by engineering an iterative consensus-based realignment strategy using BWA-MEM2, boosting alignment to 37.59% and enabling SNP calling across 10 genomic regions (LoFreq; coverage 50x, Q30 threshold).
- Assembled the viral genome de novo using SPAdes, validated with QUAST and megaBLAST; produced a 9,532 bp consensus contig with 0 ambiguous bases and confirmed homology to reference viral sequences.

Tools: FastQC, MultiQC, Trim Galore, BWA-MEM2, SAMtools, LoFreq, SPAdes, QUAST, IGV, megaBLAST

CosMx Spatial Transcriptomics Pipeline (R/Seurat)

Jul 2026

End-to-end pipeline for CosMx SMI spatial transcriptomics: SCTransform, PCA/UMAP, Louvain clustering, InSituType cell typing (8 classes, 94-99% confidence), and spatialTIME Nearest Neighbor G spatial statistics on 17,622 cells.

Tools: Scanpy, Squidpy, Python, UMAP, Leiden clustering

Machine Learning Pipeline for Biomedical Classification (Python)

Aug 2025

- Built a supervised classification pipeline (scikit-learn) applying logistic regression and random forest models with grid search hyperparameter tuning, evaluated via ROC-AUC and confusion matrix metrics.
- Applied feature importance analysis to identify the most predictive variables; workflow designed to mirror cancer genomics use cases, including patient stratification and biomarker discovery.

Tools: Python, scikit-learn, Pandas, NumPy, Matplotlib

PUBLICATIONS & PREPRINTS

Pak A, Zarei M, et al. *Artificial Intelligence, Multi-Omics, and Lung Cancer: Early Detection and Diagnostic Accuracy: A Systematic Review and Meta-Analysis.* Submitted to *Lung Cancer*; **under peer review**. Prospero registration: CRD420261417689. (2026)

SKILLS

Programming & Scripting: R (DESeq2, edgeR, ggplot2, tidyverse, Seurat, SCTransform, Bioconductor), Python (Scanpy, Squidpy, Biopython, Pandas, NumPy, Matplotlib, scikit-learn), Bash/Shell scripting, Git (version control)

AI-Assisted Productivity: Integrating large language models (Claude, GitHub Copilot) to accelerate R/Python development, debugging and applying AI-supported exploratory analysis to speed interpretation of high-dimensional omics data; prompt engineering for reproducible bioinformatics workflows; Google Colab for cloud

NGS & Omics Analysis: Bulk RNA-seq, scRNA-seq, spatial transcriptomics (CosMx SMI, InSituType cell typing, spatialTIME), ATAC-seq, ChIP-seq; differential expression, functional enrichment (GO, GSEA, MSigDB), variant calling, de novo assembly

Statistics & Machine Learning: Linear models, ANOVA, FDR correction, dimensionality reduction (PCA, UMAP, t-SNE), supervised classification (logistic regression, random forest), ROC-AUC evaluation, spatial point-pattern analysis (Nearest Neighbor G, spatial permutation testing)

Pipeline & Workflow Tools: FastQC, MultiQC, Trim Galore, BWA-MEM2, SAMtools, featureCounts, LoFreq, SPAdes, QUAST, IGV, Galaxy.eu, Metascape, ToppGene, Enrichr

Bioinformatics Databases: NCBI, Ensembl, GEO, TCGA, UniProt, KEGG, Reactome, STRING

Wet Lab Techniques: qPCR, Western blot, agarose gel electrophoresis, cell culture, flow cytometry, DNA extraction

Other: Scientific writing, data visualisation, interdisciplinary communication, Linux/Ubuntu

CERTIFICATIONS & TRAINING

Advanced Single-Cell RNA Sequencing (Nexintek)	2024
NGS Data Analysis (Nexintek)	2024
Introduction to Machine Learning (Udemy)	2023
Bioinformatics Applications in Protein Analysis (Coursera)	2022

CONFERENCES & SEMINARS

International Workshop NO-CANCER (attended)	May 2024
Diamond Light Source Early Career Scientists Symposium (attended)	Sep 2023
2nd International Conference on Biotechnology and Global Development (attended)	Apr 2021

LANGUAGES

English: Full Professional Proficiency | **Italian:** A2 (actively developing)